

Going Live with FFMPEG

FFMPEG is a free and open source project that produces libraries and programs for handling multimedia files and data. This article shows how one can upload a video file, screen cast, or Web cam output for *YouTube Live* using FFMPEG.

Sometimes, advanced technology doesn't work too well. Streaming over the Internet is an example of this. Our old bulky TV sets play live channels in real-time without any complications while a computer needs 'this and that' to stream a single video. Playing video files on a website was an ugly job for both the developer and the user till HTML5 entered the scene.

Although video and audio streaming over the Internet has been made simple, live streaming is still challenging. But that is not the case with YouTube. Uploading video for live streaming is an easy task and the user has to do nothing more than own a device that can play regular YouTube videos.

Let us look at how we, as free software enthusiasts, can broadcast on *YouTube Live*. Let's use the highly sophisticated, portable, yet simple tool called FFMPEG. The command to invoke FFMPEG can be used in other situations too—to upload videos to sites other than YouTube which support the same technology. I have taken YouTube as the example since it seems to be the most stable and popular service.

How does it work?

The basic steps involved in *YouTube Live* broadcasting are somewhat as follows.

1. You have or get a verified YouTube account, which is in *good standing*.
2. You get the public server URL and private stream name/key from your *YouTube Live* dashboard, since it's required to configure things like monetization.
3. Using these two values, you upload your video to YouTube with the help of an *encoder*.
4. The stream is broadcast live and the world gets to watch you.

RTMP and encoders

RTMP (Real Time Messaging Protocol) is an Internet protocol that can be used to stream audio, video and data between a Flash player and a server. It was initially developed as a

proprietary protocol by Macromedia. But Adobe, which owns Macromedia now, has released an incomplete version of the specification for public use.

RTMP is implemented over the following three stages:

- Live video encoder
- Live and on-demand media streaming server
- Live and on-demand media streaming client

There are free software servers and clients available to make use of RTMP. Thus you should be able to replace Adobe Flash Media Server with Red5 or FreeSWITCH, and Flash Player with Gnash or VLC. But in the case of YouTube, Google takes care of the server and client. As a YouTuber, all you have to bother about is the *encoder*.

As the broadcaster, you have to encode and upload the video to YouTube using a supported encoder. Google lists some supported encoders in its support page. There you can see an open source suggestion, *Open Broadcaster Software*. You may try it, but here we are taking up FFMPEG.



Note: The latest version of Firefox is needed to enjoy YouTube's HTML5 player. Chromium users are recommended to install or update the package *chromium-codecs-ffmpeg-extra*.

Why FFMPEG?

Because it is the simplest option! Most distros have FFMPEG preinstalled; and all you have to do is run a single line of command and you are live on YouTube!

Also, FFMPEG is highly portable (it runs on different operating systems) and powerful. It supports a lot of video formats and configurations. It can take live input from devices and push the output to remote locations. You can feel its speed even on an old computer.

It is, therefore, used as a backend of many other GUI applications, including converters and video editors. The most important thing about it is that it is a free software package!



Note: Ubuntu now ships *avconv* instead of FFMPEG.

Let's broadcast a video

Let us go through a step-by-step 'how to'. It includes both YouTube and FFMPEG.

First, you have to prepare the video file. Although

